From Overstock to On-Point: T.T. Inc's Inventory Optimization Journey

SQL Capstone Project

By

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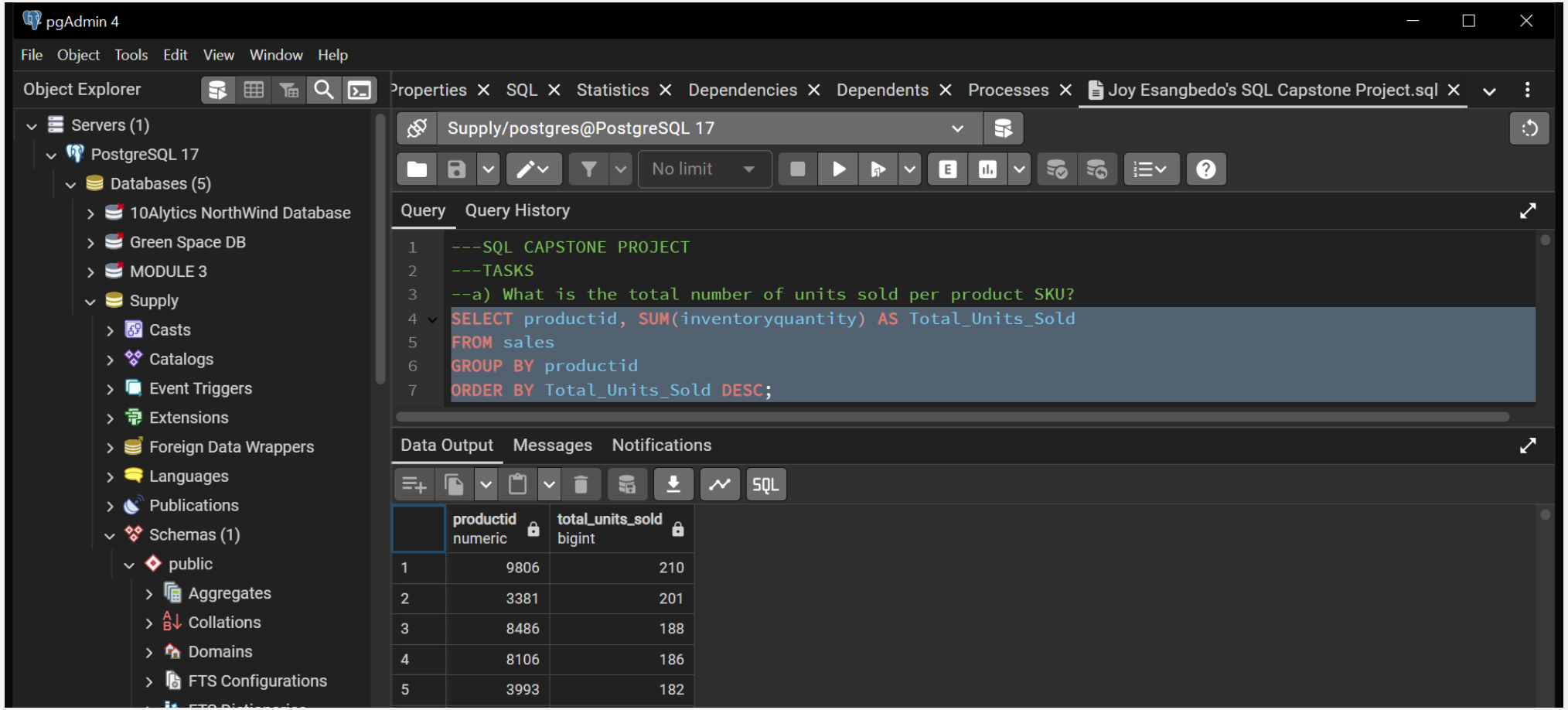
Objectives

- Highlight key findings from inventory data to identify optimization opportunities.
- Present seasonal sales trends and recommend inventory planning strategies.
- Demonstrate how inventory improvements enhance product availability and customer satisfaction.
- Provide actionable insights to support data-driven decision-making for the upcoming year's supply chain strategy.

Total Units Sold by Product SKU



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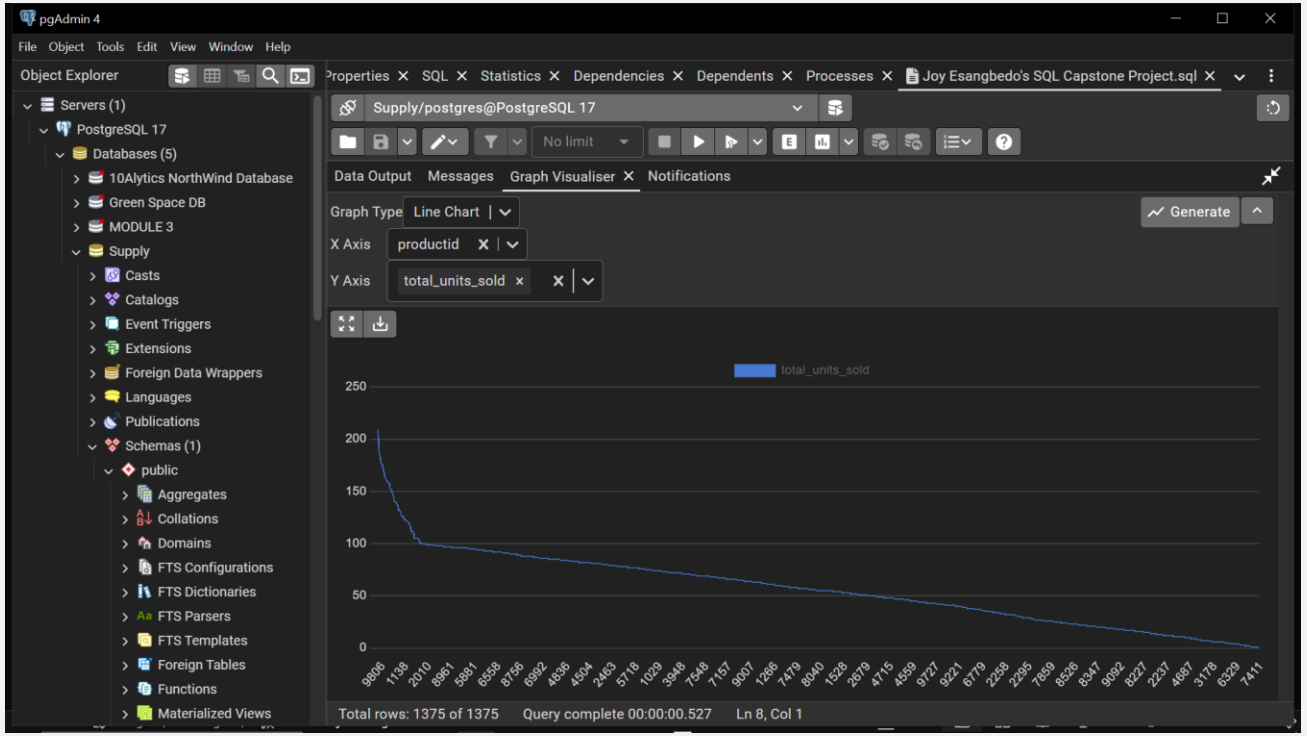
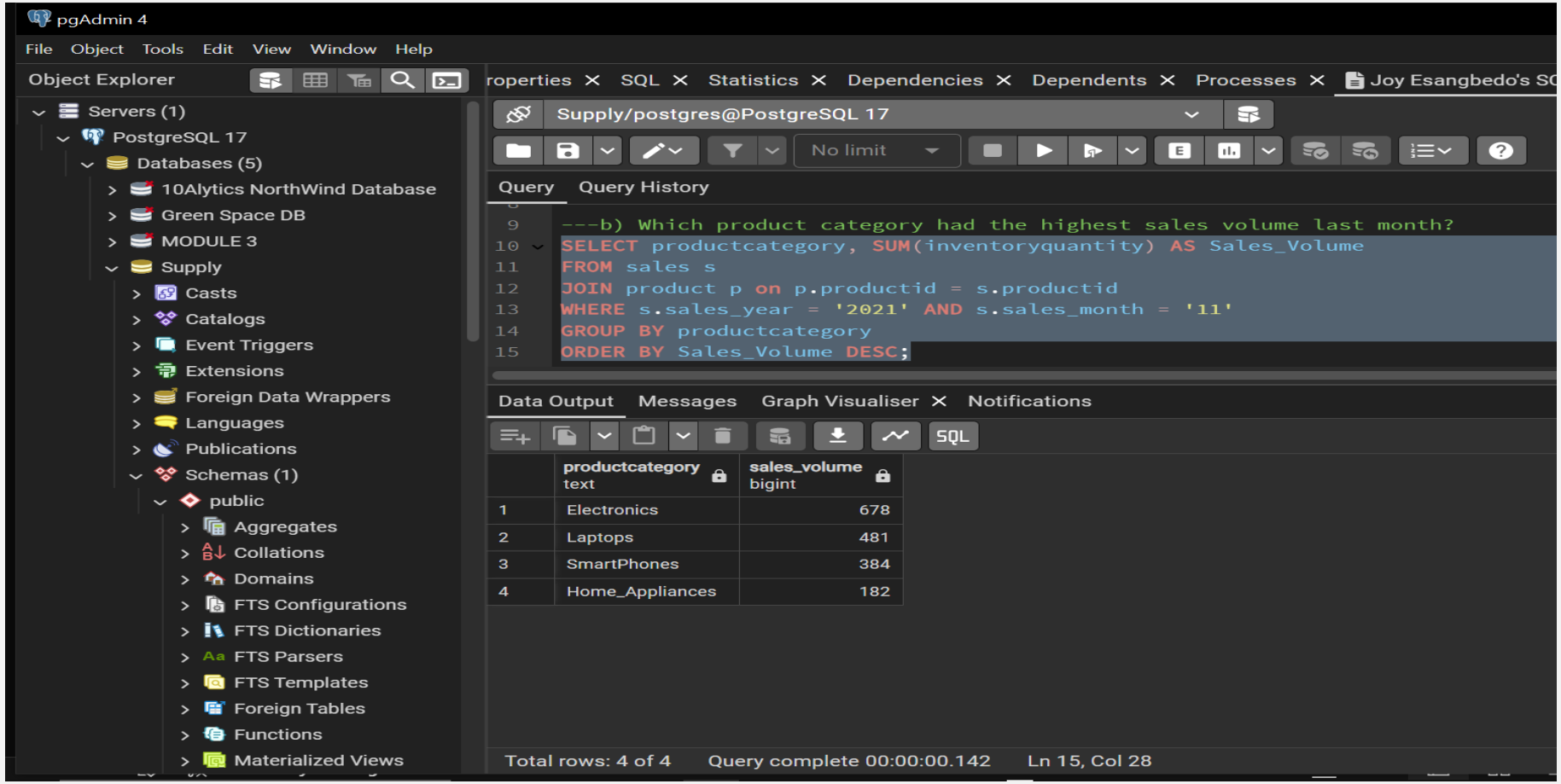
As seen in the chart:

- The productid 9806, 1138, and 2010 with the highest sales volumes should be prioritized for restocking to meet demand and maximize revenue.

- Products with lower sales may indicate overstocking risks and need for inventory adjustments or marketing efforts.



Top Sales by Product Category



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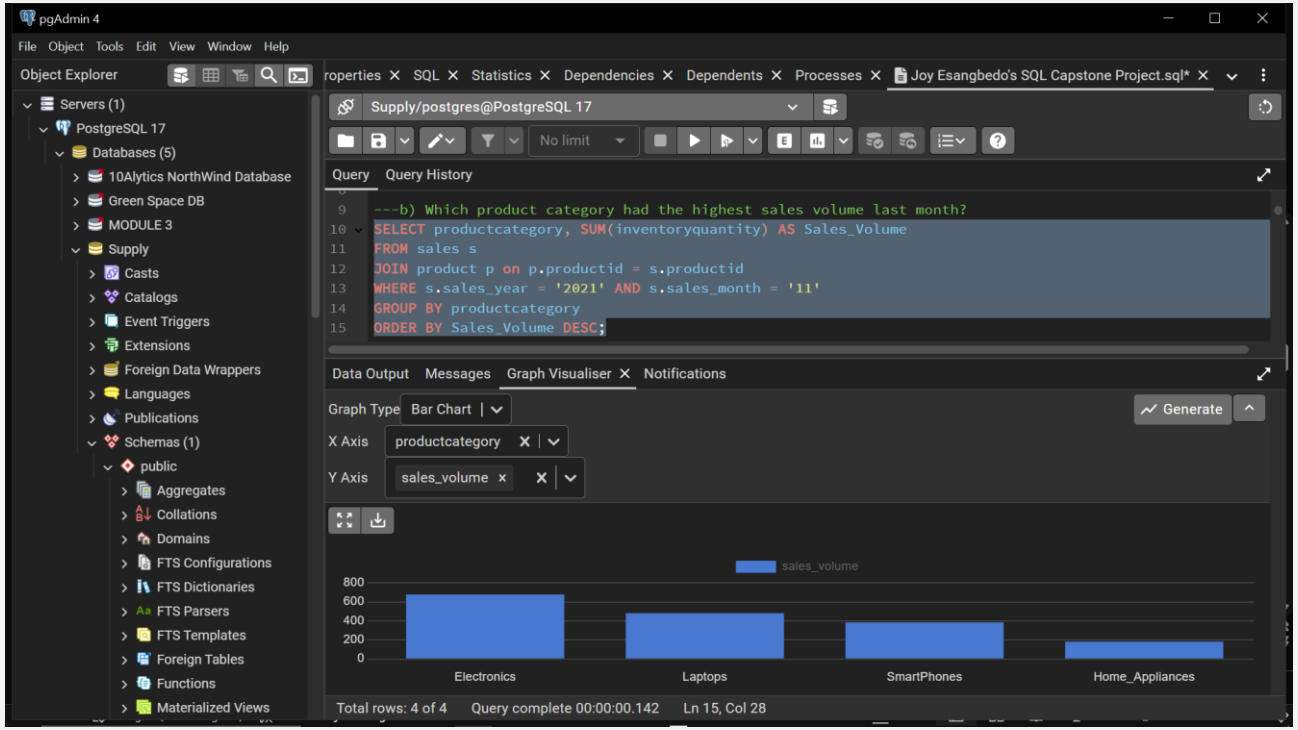
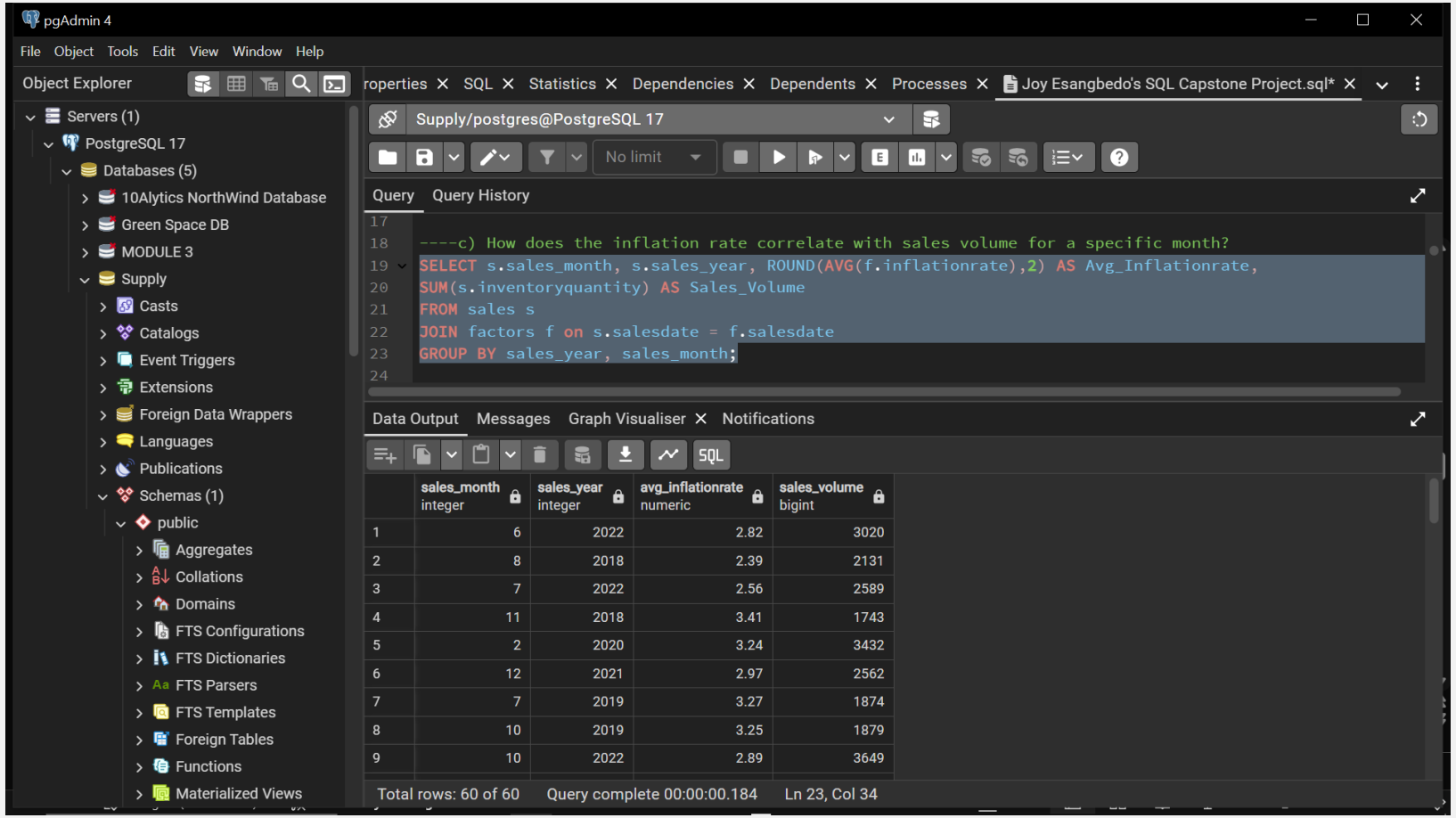
- Electronics has the highest sales volume, indicating strong demand.
- Prioritize inventory allocation for this category to prevent understocking.
- Increase safety stock levels to avoid missed sales opportunities.



Inflation Rate on Sales Volume



	sales_month integer	sales_year integer	avg_inflationrate numeric	sales_volume bigint
1	6	2022	2.82	3020
2	8	2018	2.39	2131
3	7	2022	2.56	2589
4	11	2018	3.41	1743
5	2	2020	3.24	3432
6	12	2021	2.97	2562
7	7	2019	3.27	1874
8	10	2019	3.25	1879
9	10	2022	2.89	3649

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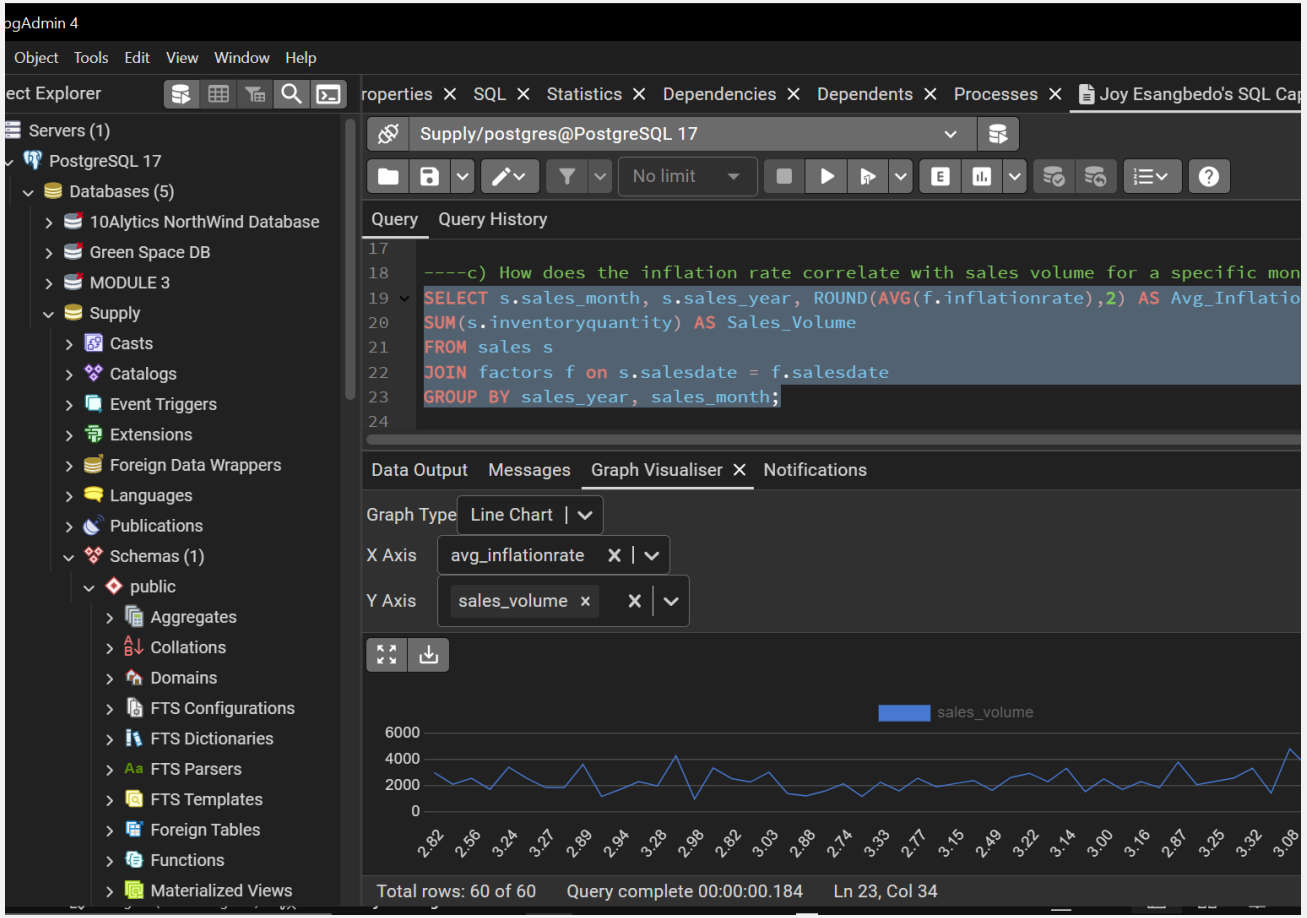
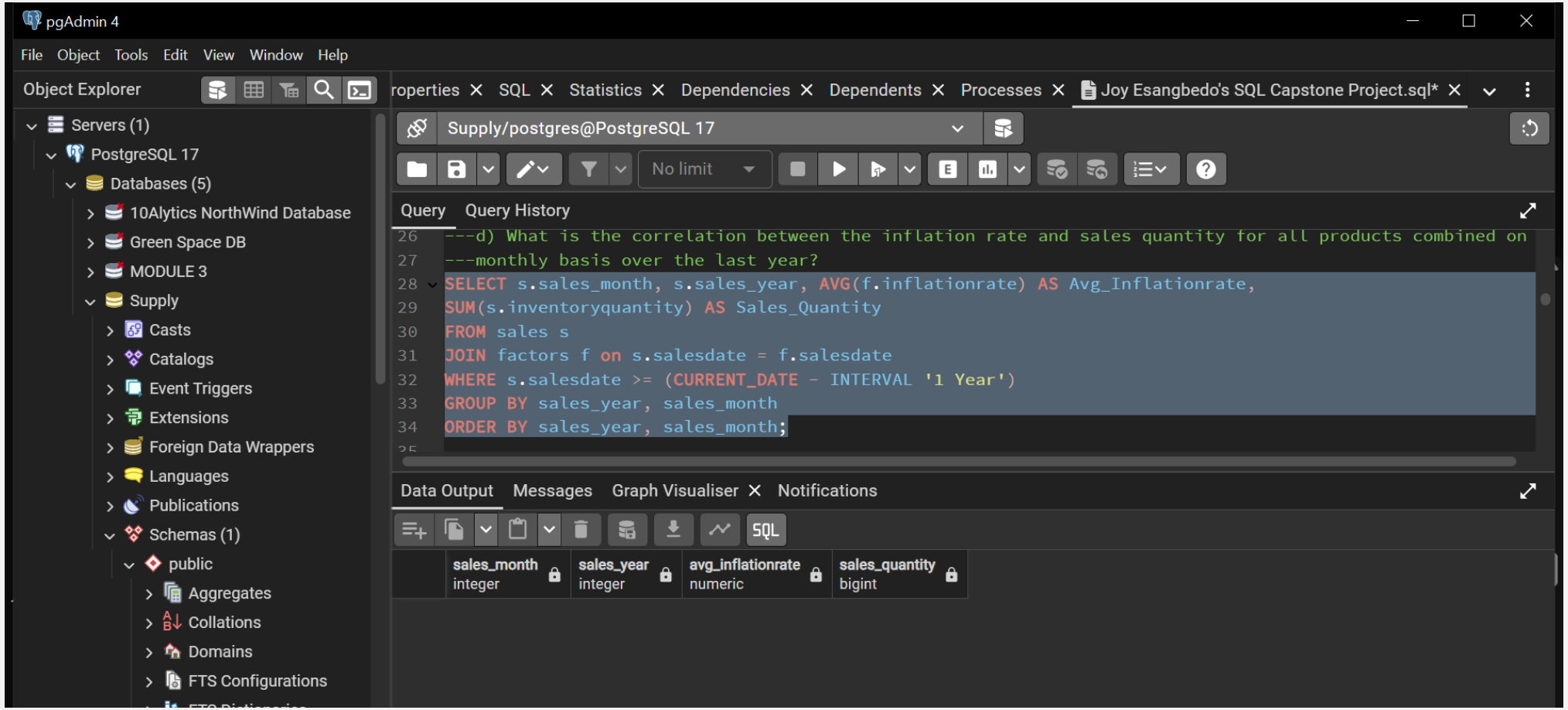
- Based on the insights it shows that the rate of inflation has no direct correlation with the sales volume.
- It suggests that other economic indices like seasonal changes could possibly have impacted sales.



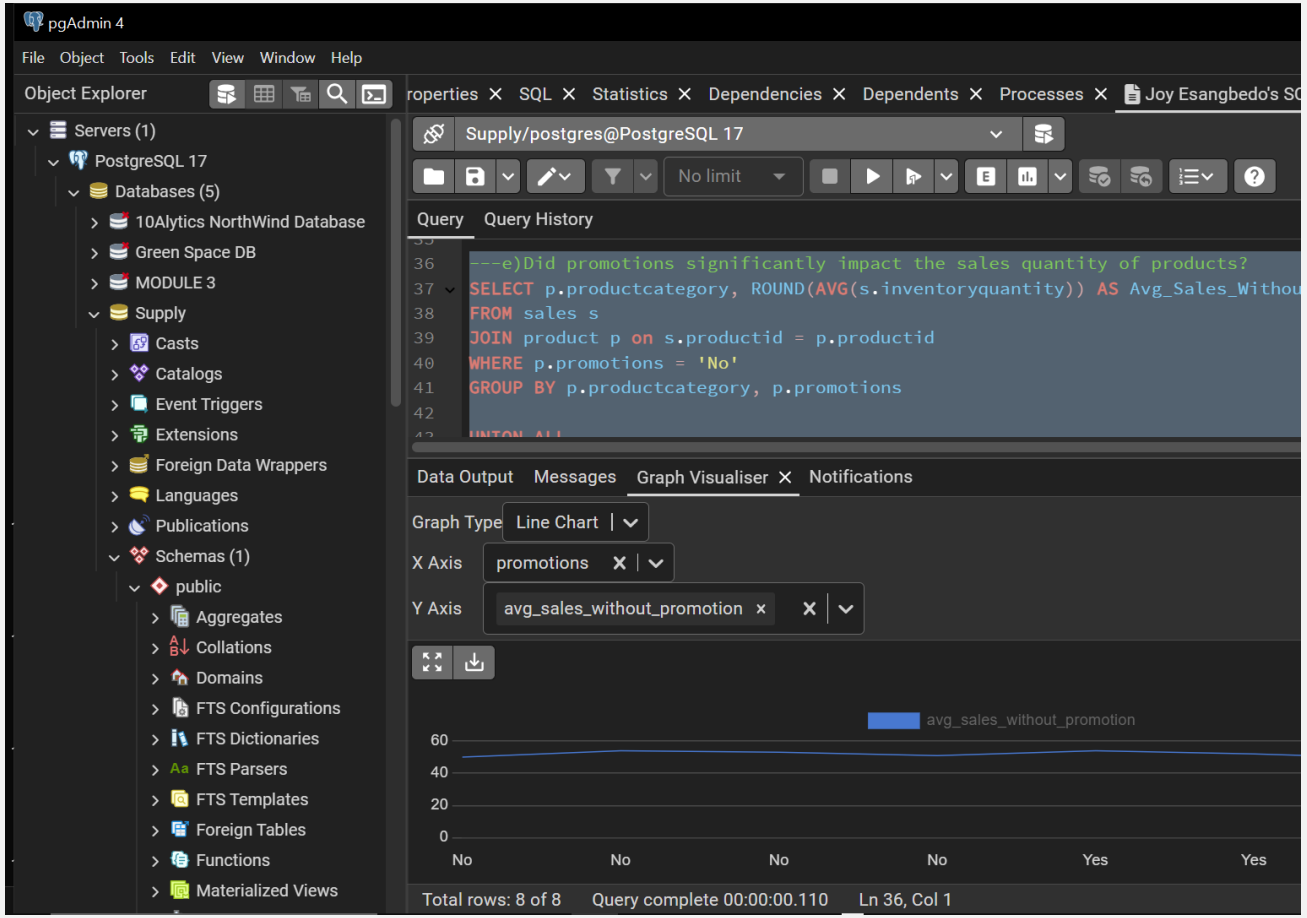
Inflation rate as per Sales on all products for each year per month



Promotion impact on Sales quantity

- The data insights showed no significant impact of promotional activities on sales quantity of the products.

- The lack of impact may indicate that current promotions do not resonate with customers or fail to target the right audience.



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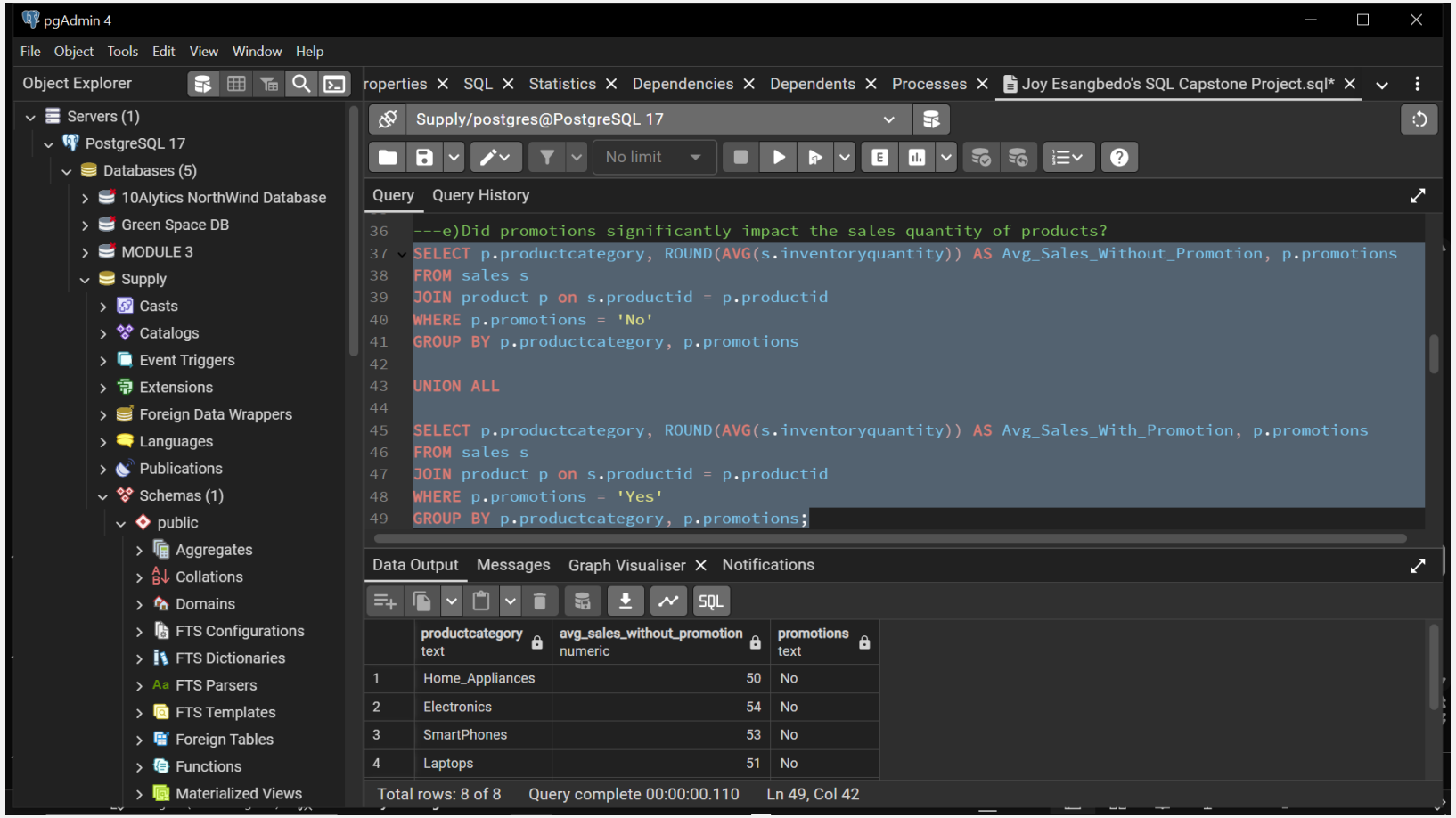
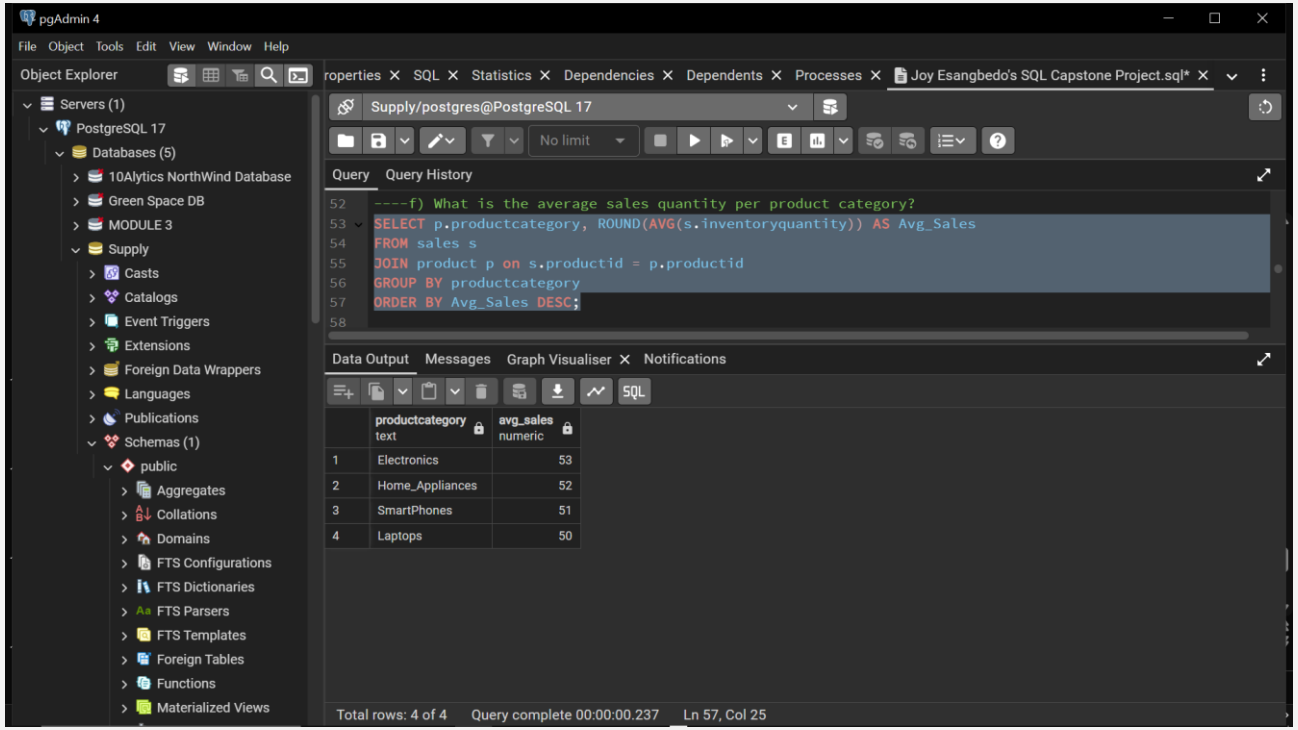

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (1)
PostgreSQL 17
Databases (5)
10Alytics NorthWind Database
Green Space DB
MODULE 3
Supply
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
roperties SQL Statistics Dependencies Dependents Processes Joy Esangbedo's SQL Capstone Project.sql*
Supply/postgres@PostgreSQL 17
No limit
Query Query History
---e)Did promotions significantly impact the sales quantity of products?
SELECT p.productcategory, ROUND(AVG(s.inventoryquantity)) AS Avg_Sales_Without_Promotion, p.promotions
FROM sales s
JOIN product p on s.productid = p.productid
WHERE p.promotions = 'No'
GROUP BY p.productcategory, p.promotions
UNION ALL
SELECT p.productcategory, ROUND(AVG(s.inventoryquantity)) AS Avg_Sales_With_Promotion, p.promotions
FROM sales s
JOIN product p on s.productid = p.productid
WHERE p.promotions = 'Yes'
GROUP BY p.productcategory, p.promotions;
Data Output Messages Graph Visualiser Notifications
SQL
productcategory text
avg_sales_without_promotion numeric
promotions text
1 Home_Appliances 50 No
2 Electronics 54 No
3 SmartPhones 53 No
4 Laptops 51 No
Total rows: 8 of 8 Query complete 00:00:00.110 Ln 49, Col 42

Average Sales per Product Category

- Electronics: Increase safety stock levels to avoid missed sales opportunities.
- Home Appliances: Implement targeted promotions or bundle strategies to move slow-moving inventory, reducing carrying costs.
- Laptops and Smartphones: Monitor sales closely and align stock levels to maintain availability without overstocking.



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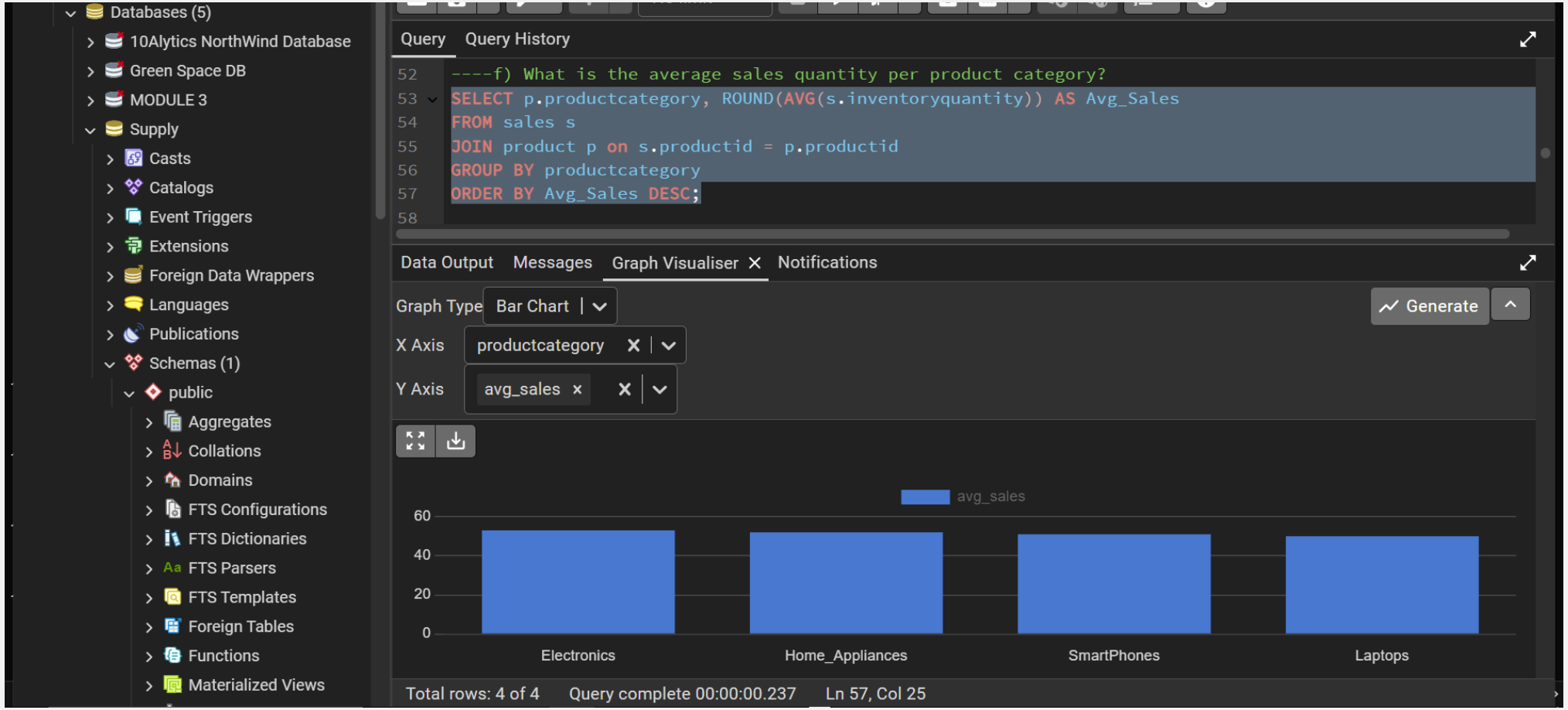
Databases (5)
10Alytics NorthWind Database
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MODULE 3
Supply
Casts
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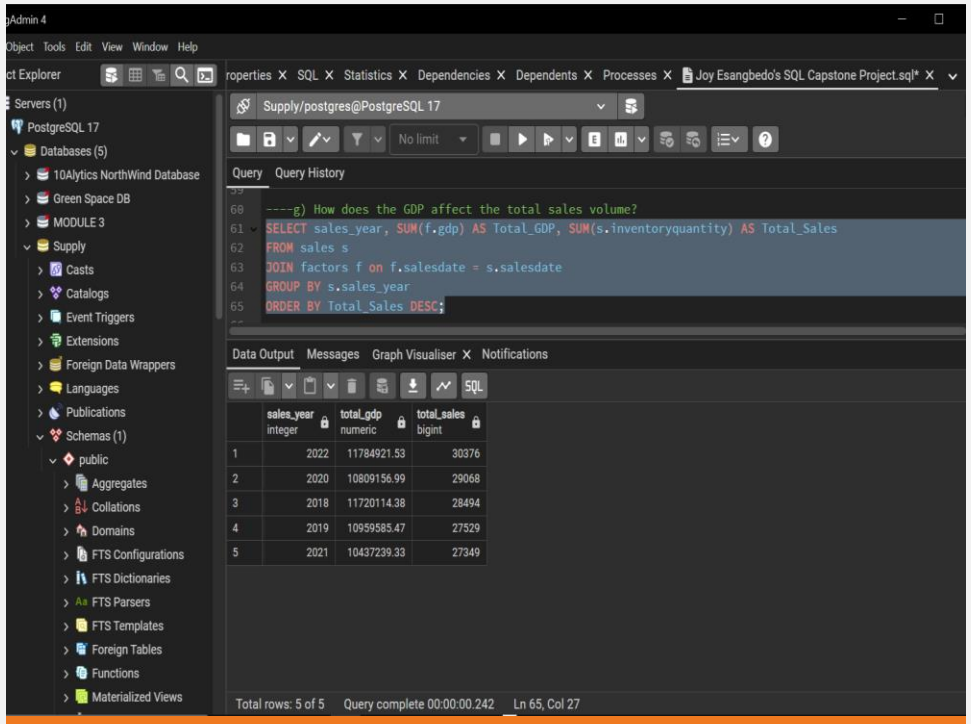
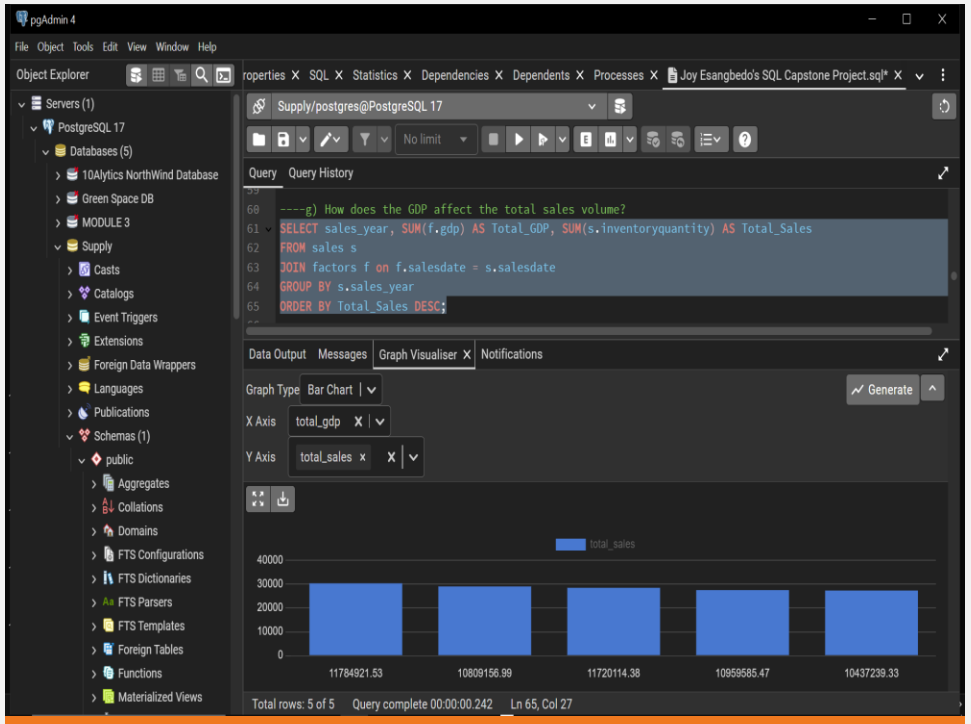
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Query
Query History
----f) What is the average sales quantity per product category?
SELECT p.productcategory, ROUND(AVG(s.inventoryquantity)) AS Avg_Sales
FROM sales s
JOIN product p on s.productid = p.productid
GROUP BY productcategory
ORDER BY Avg_Sales DESC;
Data Output
Messages
Graph Visualiser
Notifications
Graph Type
Bar Chart
Generate
X Axis
productcategory
Y Axis
avg_sales
avg_sales
60
40
20
0
Electronics
Home_Appliances
SmartPhones
Laptops
Total rows: 4 of 4
Query complete 00:00:00.237
Ln 57, Col 25

Sales Volume per GDP

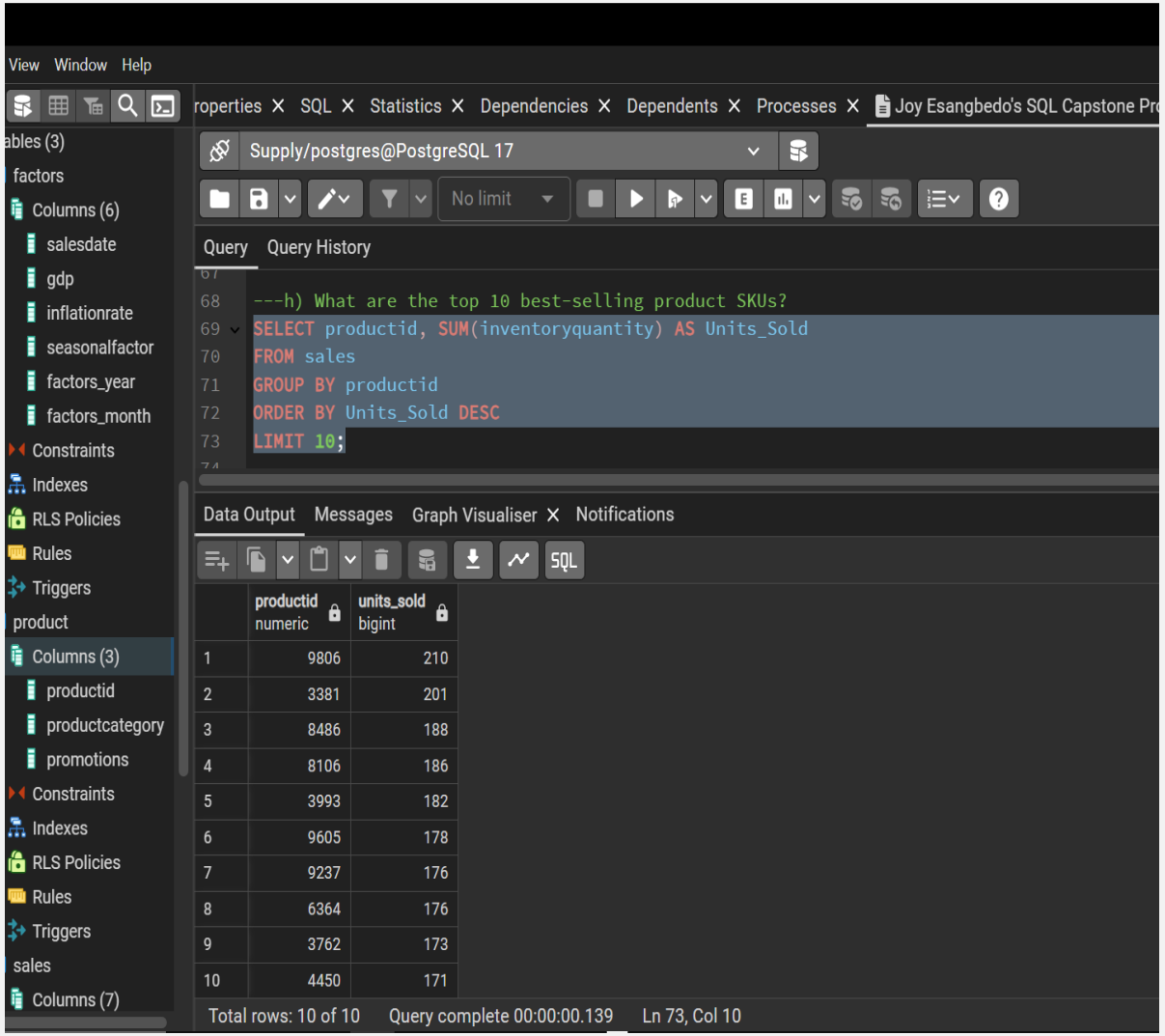


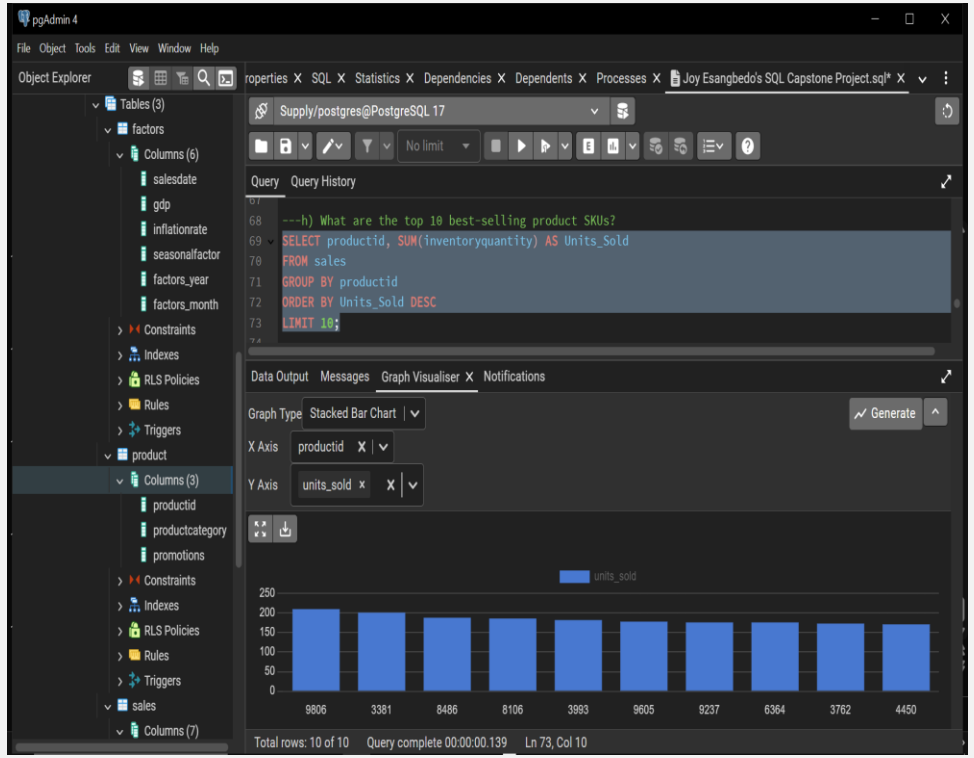
Based on the data, GDP shows fluctuations between 2018 – 2022 with sales volume remaining relatively stable.



This lack of correlation suggests that product availability may play a more significant role in sales performance.

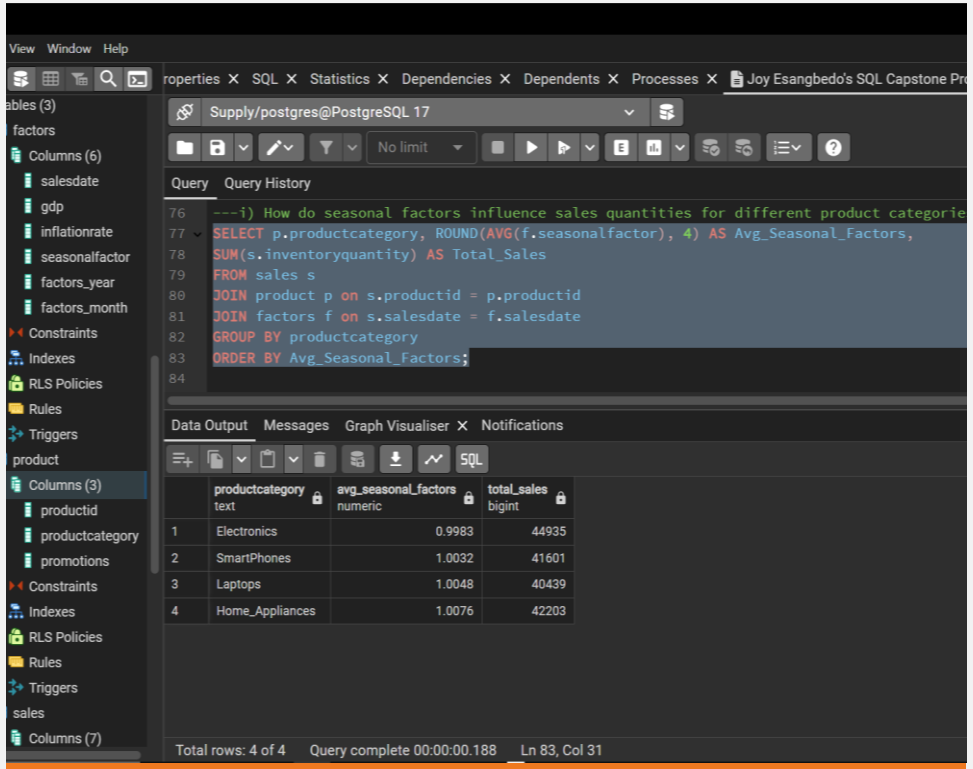
Top 10 Best Selling Products



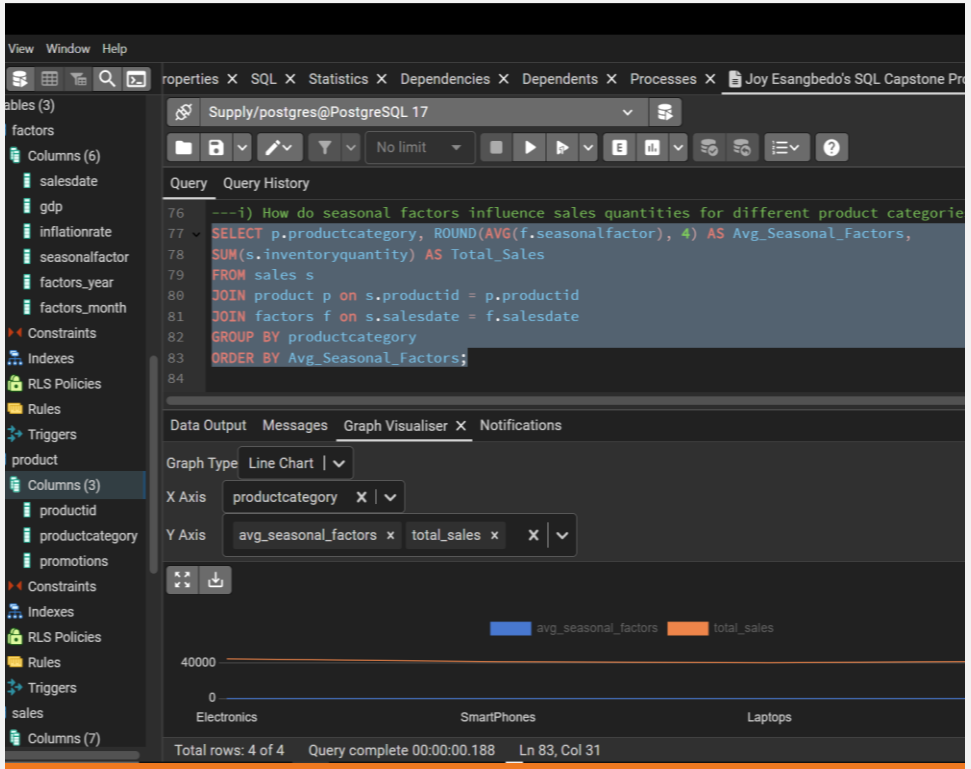


This data suggests that ensuring their availability and timely restocking is critical to avoid missed sales opportunities.

Seasonal Factors as per Sales of Product Category

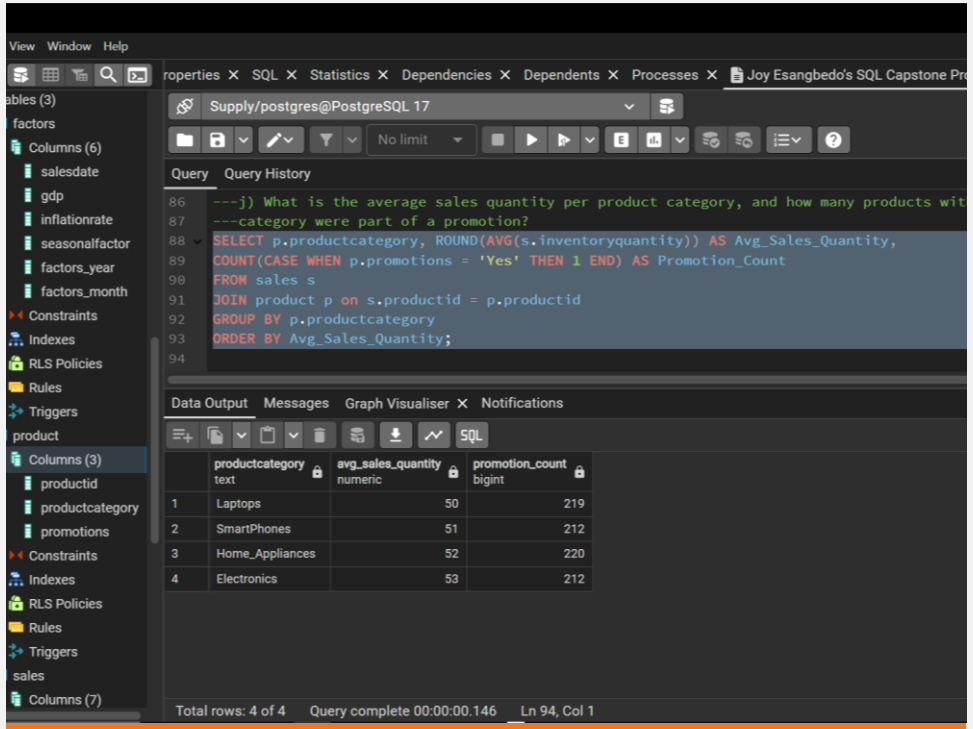


Absence of direct correlation between seasonal factors on sales quantity for product category. Customers may prioritize these products based on need rather than timing

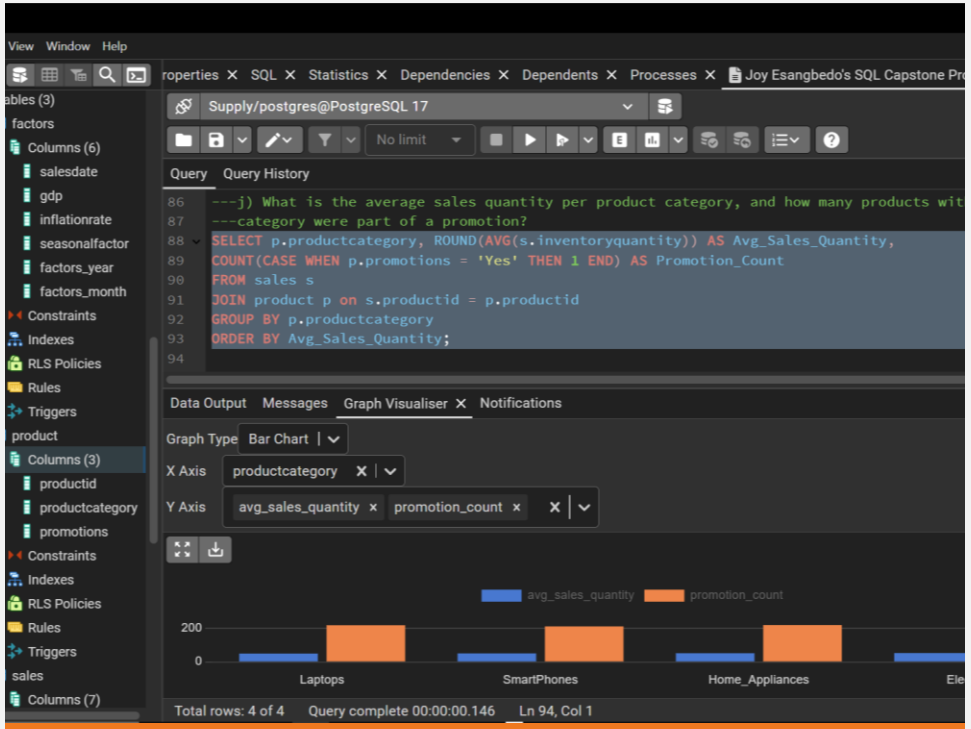


Indicates the products are either work-related (e.g., laptops) or tied to individual preferences rather than seasonal trends.

Product Category as per Avg Sales and Promotion



The average sales quantity showed a high sales quantity with low promotion of product category and vice versa



Product categories with high promotional efforts but lower sales suggest that promotions may not be effectively targeted or aligned with customer preferences.

Recommendations

Monitor key drivers beyond GDP by using predictive analytics to identify patterns and ensure proactive decision-making.

Analyze profitability by ensuring the top 10 SKUs have healthy profit margins.

Enhance post-sale services, including warranties or extended support, to boost satisfaction and loyalty.

Explore the impact of broader market trends (e.g., economic stability, customer lifestyle changes, or technological advancements) on sales performance.

Conclusion

REALLOCATE RESOURCES TO IMPACTFUL INITIATIVES

LEVERAGE CUSTOMER INSIGHTS

REFINE PROMOTIONAL TACTICS

ENHANCE SALES EFFICIENCY

MAXIMIZE ROI

IMPROVE CUSTOMER EXPECTATIONS

Thank you